# Predicting MLB Hall of Famers

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# Overview

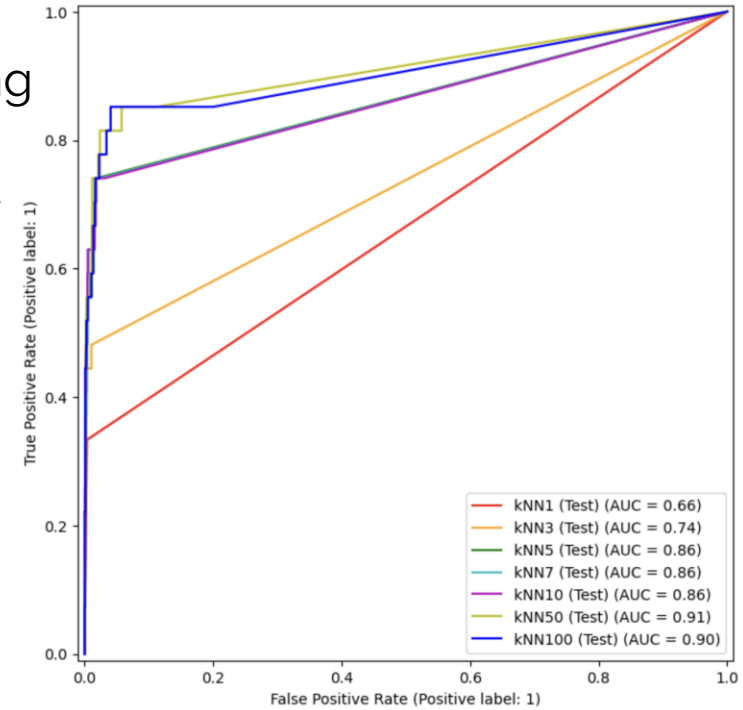
- We utilized the Lahman Baseball database which contains complete major league batting and pitching statistics from the 1871 to 2024 regular MLB seasons.
- The People, HallOfFame, Pitching, and Batting csv's were utilized.
- Each record contained statistics from one season player had its own stat line per season, but we aggregated it into career stats.
- Goal is to accurately predict which current players will make it into the MLB Hall of Fame.
- We trained on retired players and tested on both retired and active players.

# K-Nearest Neighbor

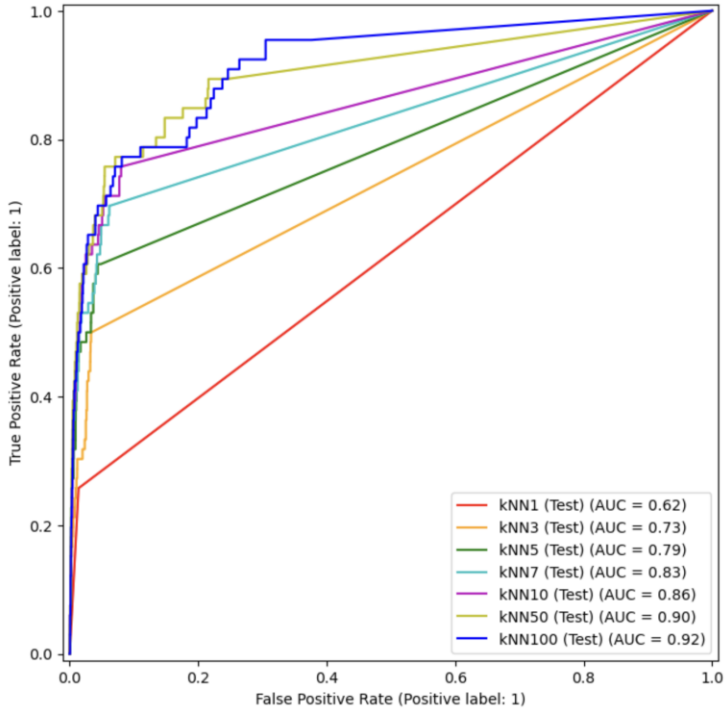
- Unable to make accurate predictions on Hall of Fame prediction
- 264 out of 20985 batters made it to the Hall of Fame. Percentage of batters that make it to the hall of fame: **1.26%**
- 107 out of 10608 pitchers made it to the Hall of Fame. Percentage of pitchers that make it to the hall of fame: **1.01%**

Pitching

kNN50
False Positive Rate: 0.0004
False Negative Rate: 0.778




| Pitching Predictions | | | Batting Predictions | | |
|---|---|---|---|---|---|
| Known Hall of Famers – knn50 wrighted model: | | | Known Hall of Famers – knn7 weighted model: | | |
| Bob Gibson | 100.0% | ✓ HOF | Hank Aaron | 100.0% | ✓ HOF |
| Roy Halladay | 100.0% | ✓ HOF | Derek Jeter | 100.0% | ✓ HOF |
| Walter Johnson | 100.0% | ✓ HOF | Chipper Jones | 100.0% | ✓ HOF |
| Nolan Ryan | 100.0% | ✓ HOF | Willie Mays | 62.0% | ✓ HOF |
| Mike Mussina | 32.7% | ✗ Not HOF | Babe Ruth | 59.2% | ✓ HOF |
| Active Players – knn50 weighted model: | | | Active Players – knn7 weighted model: | | |
| Max Scherzer | 25.5% | ✗ Unlikely HOF | Miguel Cabrera | 0.0% | ✗ Unlikely HOF |
| Craig Kimbrel | 5.5% | ✗ Unlikely HOF | Nelson Cruz | 0.0% | ✗ Unlikely HOF |
| Kenley Jansen | 0.0% | ✗ Unlikely HOF | Freddie Freeman | 0.0% | ✗ Unlikely HOF |
| Clayton Kershaw | 0.0% | ✗ Unlikely HOF | Andrew McCutchen | 0.0% | ✗ Unlikely HOF |
| Justin Verlander | 0.0% | ✗ Unlikely HOF | Joey Votto | 0.0% | ✗ Unlikely HOF |




kNN100
False Positive Rate: 0
False Negative Rate: 1

Batting

# Pitching Random Forest

Data Preprocessing:
- Added ERA_career, WHIP, IP, HR9, BB9, and K9 as features
- Filtered out position players who had only pitched in a blowout

- The top features in the feature importance list include wins, strikeouts, career ERA (earned run average), and innings pitched
- This suggests that a long career with high performance matters most

| | |
|---|---|
| W | 0.132052 |
| SO | 0.126738 |
| ERA_career | 0.088071 |
| IP | 0.085296 |
| WHIP | 0.076488 |
| H | 0.072482 |
| G | 0.072325 |
| GS | 0.061929 |
| L | 0.053819 |
| HR9 | 0.048910 |
| BB | 0.046195 |
| BB9 | 0.039765 |
| SV | 0.032422 |
| HR | 0.032287 |
| K9 | 0.031220 |

Feature importance list from random forest

- Accuracy (retired players): 0.92
- False positive rate: 0.08
- False negative rate: 0.0
- ROC-AUC: 0.981

Random Forest Top 20 Players

| nameGiven | nameLast | hof_prob |
|---|---|---|
| Clayton Edward | Kershaw | 0.427257 |
| Justin Brooks | Verlander | 0.401765 |
| Kenley Geronimo | Jansen | 0.330822 |
| Maxwell Martin | Scherzer | 0.307615 |
| Craig Michael | Kimbrel | 0.237924 |
| Albertin Aroldis | Chapman | 0.174006 |
| David Alan | Robertson | 0.116375 |
| Christopher | Sale | 0.104619 |
| Jacob Anthony | deGrom | 0.094223 |
| Gerrit Alan | Cole | 0.084606 |
| Joseph Peter | La Sorsa | 0.026590 |
| Kyle Yandow | Leahy | 0.024693 |
| Dylan Lee | Floro | 0.023619 |
| Noah D. | Davis | 0.021786 |
| Devin Terran | Williams | 0.021016 |
| Zachary Peter | Gallen | 0.017832 |
| Zachary Harrison | Wheeler | 0.017491 |
| Edwin Orlando | Diaz | 0.013396 |
| Blake Ashton | Snell | 0.010926 |
| Paul David | Skenes | 0.009716 |

# Pitching Logistic Regression

- 1st Logistic Regression model containing every feature – same used in random forest model
- Accuracy: 0.85
- False Positive Rate: 0.15
- False Negative Rate: 0.26
- ROC-AUC: 0.81
- Some out of place players which don't really make sense – ex: Enoli Paredes who played 4 seasons with weak performance

| nameGiven | nameLast | hof_prob_logistic |
|---|---|---|
| Craig Michael | Kimbrel | 0.974908 |
| Justin Brooks | Verlander | 0.943061 |
| Enoli Norverto | Paredes | 0.937781 |
| Albertin Aroldis | Chapman | 0.933586 |
| Kenley Geronimo | Jansen | 0.925669 |
| Maxwell Martin | Scherzer | 0.882061 |
| Clayton Edward | Kershaw | 0.860509 |
| Devin Terran | Williams | 0.651739 |
| Alexis Omar | Diaz | 0.586482 |
| Josh Ronald | Hader | 0.517883 |
| Joseph Patrick | Boyle | 0.457628 |
| Gerrit Alan | Cole | 0.429891 |
| Jake Owen | Cousins | 0.423120 |
| Christopher | Sale | 0.407046 |
| Edwin Orlando | Diaz | 0.334421 |
| Justin | Martinez | 0.324056 |
| Camilo | Doval | 0.306585 |
| Landen Brice | Roupp | 0.299122 |
| Ryan Dalton | Helsley | 0.283254 |
| Dylan David | Coleman | 0.264215 |

- 2nd Logistic Regression model containing top 5 features: wins, strikeouts, career ERA, innings pitched, and WHIP
- Accuracy: 0.83
- False Positive Rate: 0.17
- False Negative Rate: 0.26
- ROC-AUC: 0.86
- Top players make more sense as opposed to the previous model – Justin Verlander, Clayton Kershaw, Max Scherzer

| nameGiven | nameLast | hof_prob_logistic2 |
|---|---|---|
| Justin Brooks | Verlander | 0.966564 |
| Clayton Edward | Kershaw | 0.960346 |
| Maxwell Martin | Scherzer | 0.949065 |
| Gerrit Alan | Cole | 0.775051 |
| Christopher | Sale | 0.753086 |
| Michael Lance | Lynn | 0.637233 |
| Jacob Anthony | deGrom | 0.565591 |
| Johnny | Cueto | 0.555528 |
| Charles Alfred | Morton | 0.520505 |
| Albertin Aroldis | Chapman | 0.510108 |
| Blake Ashton | Snell | 0.471667 |
| Sefat Farid Yu | Darvish | 0.467666 |
| Sonny Douglas | Gray | 0.462516 |
| David Alan | Robertson | 0.462425 |
| Devin Terran | Williams | 0.462216 |
| Zachary Harrison | Wheeler | 0.460730 |
| Craig Michael | Kimbrel | 0.441112 |
| Kevin John | Gausman | 0.372603 |
| Max Dorian | Fried | 0.370667 |
| Aaron Michael | Nola | 0.355031 |

# Batting

## Random Forest

Data Preprocessing:

- Aggregated data to be one row per player per season
- Active/Retired split to train on retired, test on active and retired
- Filtering features to only get numeric, relevant stats
- Made sure to balance class due to a highly imbalanced dataset

### Feature Importance:

The model found that the five most important features are Number of Games Played, At-Bats, Strikeouts, Hits, Runs, and RBIs

This list makes sense, as longevity is strongly related to HOF eligibility (Shown through Games, At-bats, Runs, Hits, RBIs). Strikeouts show up as high, even though they are not necessarily good, they are predictive.

| | |
|---|---|
| G | 0.239513 |
| AB | 0.149897 |
| SO | 0.111409 |
| H | 0.099991 |
| R | 0.086894 |
| RBI | 0.067279 |
| BB | 0.052186 |
| 2B | 0.047340 |
| SH | 0.038665 |
| 3B | 0.020137 |
| HBP | 0.018720 |
| HR | 0.018195 |
| CS | 0.015485 |
| IBB | 0.015084 |
| SB | 0.011442 |
| SF | 0.007762 |

Accuracy: 0.96
False Positive Rate: 0.039
False Negative Rate: 0.029
ROC-AUC: 0.98

Initially the model had a FNR way to high, so we needed an optimal threshold, found by TPR-FPR because this dataset is highly imbalanced

After creating an optimal threshold of 0.037, these were the only players the model enough was confident enough to say would be inducted

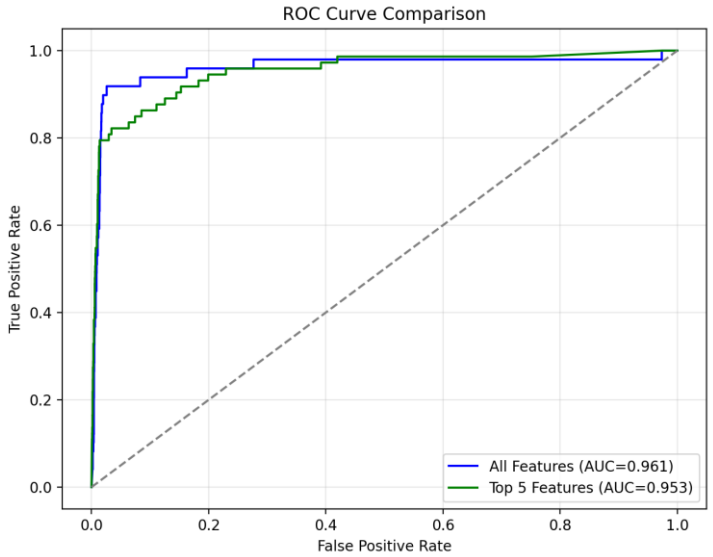
### Random Forest Top Players:

| | full_name | HOF_prob |
|---|---|---|
| 644 | Caleb Hamilton | 0.186180 |
| 1147 | Tyler Naquin | 0.130799 |
| 705 | Brewer Hicklen | 0.100857 |
| 933 | Irving Lopez | 0.093703 |
| 1442 | Jonathan Schoop | 0.039225 |
| 1453 | Corey Seager | 0.038860 |

# Batting Logistic Regression

| full_name | HOF_prob |
|---|---|
| Miguel Cabrera | 0.964428 |
| Paul Goldschmidt | 0.820179 |
| Freddie Freeman | 0.755512 |
| Bryce Harper | 0.754709 |
| Evan Longoria | 0.671471 |
| Joey Votto | 0.617013 |
| Mike Trout | 0.607713 |
| Nelson Cruz | 0.589706 |
| Giancarlo Stanton | 0.587729 |
| J. D. Martinez | 0.585880 |
| Christian Yelich | 0.533223 |
| Brandon Crawford | 0.525383 |
| Andrew McCutchen | 0.496317 |
| Matt Carpenter | 0.486509 |
| Kenley Jansen | 0.481369 |
| Aaron Judge | 0.479569 |
| Manny Machado | 0.478475 |
| David Robertson | 0.474339 |
| Eric Hosmer | 0.474204 |
| Nolan Arenado | 0.461662 |

- 1st Logistic Regression model containing every feature
- Accuracy: 0.97
  False Positive Rate: 0.10
  False Negative Rate: 0.03
  ROC-AUC: 0.961
- These predictions make sense, they reward players who are commonly regarded among fans as “the best”




- 2nd Logistic Regression model containing top 5 features: Hits, RBIs, Runs, Home Runs, and Walks
- Accuracy: 0.97
  False Positive Rate: 0.024
  False Negative Rate: 0.21
  ROC-AUC: 0.953
- These predictions seem slightly less realistic, there are some names that should not be there, and some names that should appear much higher

| full_name | HOF_prob |
|---|---|
| Miguel Cabrera | 0.833163 |
| Freddie Freeman | 0.648580 |
| Nelson Cruz | 0.621097 |
| Andrew McCutchen | 0.607154 |
| Joey Votto | 0.605442 |
| Paul Goldschmidt | 0.604334 |
| Evan Longoria | 0.593422 |
| Jose Altuve | 0.590264 |
| Elvis Andrus | 0.588111 |
| Nolan Arenado | 0.576125 |
| Manny Machado | 0.567893 |
| J. D. Martinez | 0.551514 |
| Carlos Santana | 0.546342 |
| Eric Hosmer | 0.546304 |
| Charlie Blackmon | 0.523822 |
| Jose Abreu | 0.522760 |
| Salvador Perez | 0.515553 |
| Anthony Rizzo | 0.513786 |
| Xander Bogaerts | 0.512714 |
| Michael Brantley | 0.509689 |

# Thank You